

The 21 cm hydrogen line

Reading and calculation

Use neutral hydrogen emission to understand what the horn telescope measures. Then calculate the energy of one radio photon.

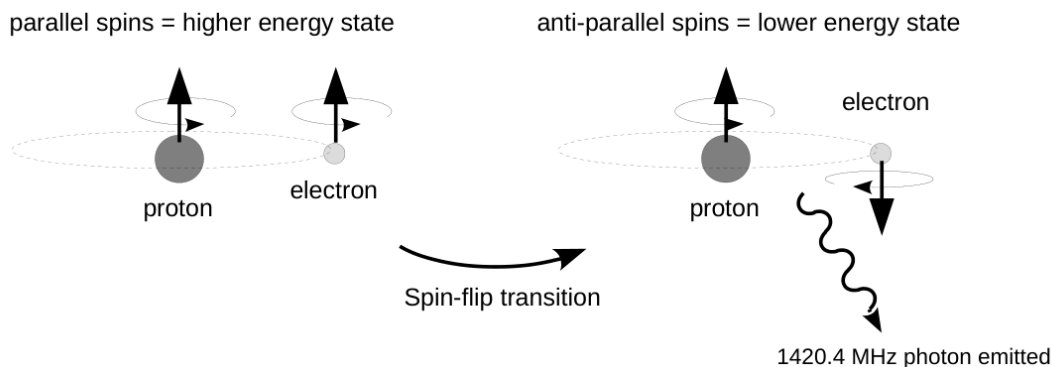
A radio view of the Milky Way

A neutral hydrogen atom has one proton and one electron. Hydrogen gas fills much of the space between stars. Its radio emission passes through dust that blocks visible light. We use this emission to map gas across the Milky Way.

How the line forms

The proton and electron have a quantum property called spin. Spin is related to angular momentum; it does not mean literal rotation. Their relative spin orientations give neutral hydrogen two closely spaced ground-state energy levels.

Parallel spins correspond to the higher energy level. Antiparallel spins correspond to the lower level. A transition to the lower level releases a photon near 1420.4 MHz, with a wavelength of about 21 cm. Absorbing a photon at this frequency can drive the reverse transition.



Hydrogen releases a radio photon when it changes to the lower energy state.

The horn telescope measures signals near this frequency. Motion along our line of sight shifts the observed frequency through the Doppler effect.

Calculate the photon energy

Use $E = hf$, where E is photon energy, h is Planck's constant, and f is frequency.

1. Look up Planck's constant in joule-seconds. Convert 1420.4 MHz to hertz.
2. Calculate the photon energy in joules. Show your units and working.
